

Batch: A3 Roll No.: 16010324044
Experiment No. 08
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: Output characteristics of an n-channel MOSFET

OBJECTIVE: To analyze the output characteristics of an n-channel MOSFET by plotting the drain current (I_D) versus the drain-to-source voltage (V_{DS}) for different gate-to-source voltages (V_G), design fixed bias and self-bias circuits of MOSFET

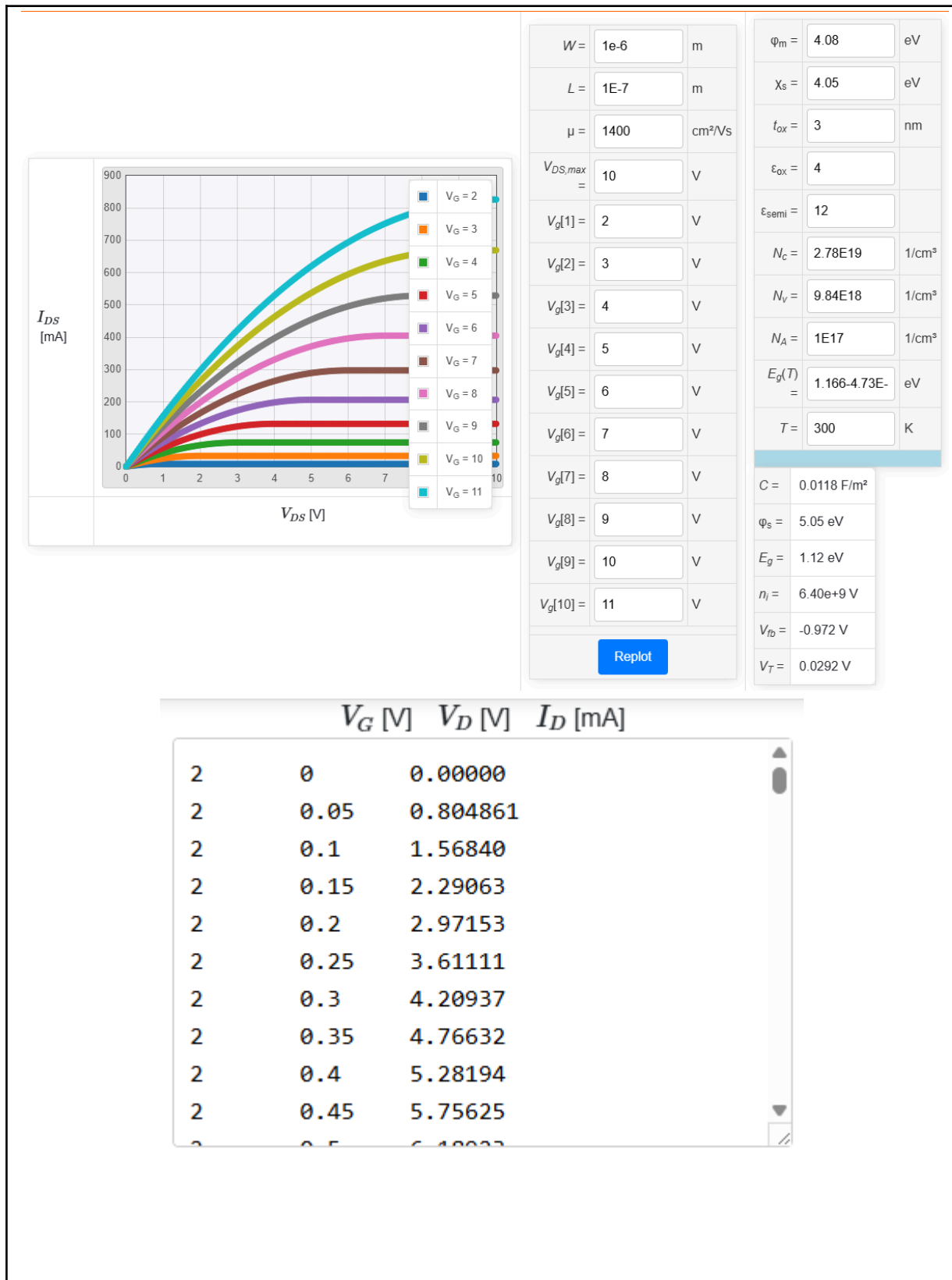
OUTCOME: Student will be able to analyze the transistor circuits for DC operation

Procedure:

1. Set the MOSFET Parameters:
 - a. Adjust the width (W) and length (L) of the MOSFET channel using the input fields. For example, set $W = 1E-6$ m and $L = 1E-7$ m.
 - b. Set the mobility (μ) value. In this case, use $\mu = 1400$ cm²/V•s.
 - c. Set the maximum drain-to-source voltage ($V_{DS,max}$) to 10 V.
2. Configure the Gate-to-Source Voltages (V_{GS}):
 - a. Enter the desired V_{GS} values in the input fields. You can set multiple values, for example, from 2 V to 11 V in steps of 1 V.
3. Set the Material Properties:
 - a. Specify the material parameters such as the work function (ϕ_m), electron affinity (χ_s), and oxide thickness (t_{ox}), among others.
 - b. Use typical values like $\phi_m = 4.08$ eV, $\chi_s = 4.05$ eV, $t_{ox} = 3$ nm, and $\epsilon_{ox} = 4$.
 - c. Set semiconductor parameters such as ϵ_{semi} (permittivity of the semiconductor), NA (acceptor doping concentration), and temperature (T).
4. Click on the "Replot" Button:
 - a. Press the "Replot" button to generate the output characteristics curves, which plot I_D (drain current) versus V_{DS} for each selected V_{GS} .
5. Analyze the Output Characteristics:
 - a. Observe how the drain current (I_D) varies with the drain-to-source voltage (V_{DS}) for different gate voltages (V_{GS}).
 - b. Identify the different operating regions (cutoff, linear, and saturation) for each curve.
6. Optional - Modify Parameters:
 - a. To further explore the MOSFET behavior, modify the parameters (e.g., W , L , V_{GS} values) and click "Replot" again to observe the changes in the output characteristics.

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VLAB screen shots:



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Post lab questions:

1. **What is the significance of the transconductance (g_m) in MOSFET operation?**
 - a: It indicates the output voltage
 - b: It indicates the input capacitance
 - c: It indicates the change in drain current with gate voltage
 - d: It indicates the change in source current with drain voltage

2. **In terms of output characteristics, what does the term 'pinch-off' refer to?**
 - a: The maximum voltage across the MOSFET
 - b: The point where the channel is fully formed
 - c: The point at which the current becomes constant despite increasing V_{DS}
 - d: The minimum gate voltage needed to turn on the device

Conclusion:

The MOSFET output curves show that the drain current increases with gate voltage, confirming it as a voltage-controlled device. The distinct cutoff, linear, and saturation regions align with the expected transistor behavior.

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